File permissions in Linux

Project description

The research team at my organization needed to update the file permissions for specific files and directories within the `projects` directory. The existing permissions did not match the organization's authorization policy. To improve security, I reviewed the current permissions, identified unauthorized access, and modified the permissions using Linux commands.

Check file and directory details

I used the following command to display all files, directories, and their permissions:

ls -la

This command displayed the contents of the `projects` directory, including hidden files, and showed the permission settings for each file and directory. From the output, I identified the permissions that needed to be changed.

Describe the permissions string

Each file and directory has a 10-character permission string.

- The **first character** indicates the file type:
    - `-` = regular file
    - `d` = directory
- The **next three characters** represent the owner's permissions (read, write, execute).
- The **following three characters** represent the group's permissions.
- The **last three characters** represent the permissions for all other users.

For example, the permission string `-rw-rw-r--` indicates:

- The item is a regular file.
- The owner has read and write permissions.

- The group has read and write permissions.
- Others have read-only permission.

Change file permissions

The organization required that **other users should not have write access** to project files. The file `project_k.txt` allowed write access for others, so I removed that permission using:

chmod o-w project_k.txt

This changed the permissions so that others have read-only access.

Change file permissions on a hidden file

The archived file `.project_x.txt` should allow only the user and group to read it, with no write permissions.

I updated the permissions using:

chmod u-w .project_x.txt
chmod g-w,g+r .project_x.txt

These commands removed write permission from both the user and group while ensuring the group retained read permission.

Change directory permissions

The organization wanted only the owner (`researcher2`) to have access to the `drafts` directory. The group previously had execute permission, so I removed it using:

chmod g-x drafts

After this change, only the owner retained execute permission for the directory.

Summary

I reviewed the permissions of files and directories in the `projects` directory using the `ls -la` command and modified them with the `chmod` command. I removed unauthorized write access from `project_k.txt`, updated the permissions of the hidden file `.project_x.txt` to make it read-only for authorized users, and restricted access to the `drafts` directory so only the owner has execute permission. These changes ensured the file system permissions complied with the organization's security requirements.